

SMS in CHC

Abstract

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1 Introduction

2 Gauge Boson Signal

2.1 Preliminaries

We will compute the connected four-point function $\langle \delta h_{\mathbf{k}_1} \delta h_{\mathbf{k}_2} \delta h_{\mathbf{k}_3} \delta h_{\mathbf{k}_4} \rangle'$ generated by a massive gauge-boson loop. Our interest is the leading nonlocal signal in the collapsed s channel,

$$\mathbf{k}_s \equiv \mathbf{k}_1 + \mathbf{k}_2 = -(\mathbf{k}_3 + \mathbf{k}_4), \quad k_s \ll k_i, \quad k_{12} \equiv k_1 + k_2, \quad k_{34} \equiv k_3 + k_4, \quad (1)$$

where $k_i \equiv |\mathbf{k}_i|$. Thus $k_1 \simeq k_2$ and $k_3 \simeq k_4$ at leading order. We choose $\mathbf{k}_s$ along the z axis and the azimuth of $\mathbf{k}_1$ to be zero, and θ_1 and θ_3 are the polar angles of $\mathbf{k}_1$ and $\mathbf{k}_3$, and ϕ_3 is their relative azimuth.

The Higgs–gauge couplings follow from the Higgs kinetic term [1]:

$$\frac{\sqrt{-g}}{2} |D_\mu \mathbf{H}|^2 \supset \frac{a^2 g^2}{4} (h_0 + \delta h)^2 \left(W_\mu^+ W^{-\mu} + \frac{Z_\mu Z^\mu}{2c_W^2} \right), \quad (2)$$

where g is the weak gauge coupling, $c_W = \cos \theta_W$, and the Lorentz contractions on the right use the flat conformal metric. The terms linear and quadratic in δh generate the cubic and quartic vertices, respectively. They give the bubble, triangle, and box topologies considered below. In the light-Higgs regime, the dominant infrared Higgs-loop contribution to the vector masses is [1]

$$m_W^2 \simeq \frac{3g^2 H^4}{8\pi^2 m_H^2}, \quad m_Z^2 \simeq \frac{m_W^2}{c_W^2}. \quad (3)$$

Here m_H is the effective Higgs mass. We use a constant vector mass m_V and the parameter $\tilde{\nu} \equiv \sqrt{m_V^2/H^2 - 1/4} > 0$ in the following. The W and Z contributions are obtained with their respective masses and couplings in (2).

We use $\lambda, \sigma = L, +, -, T$ for vector polarization-basis labels, and $\eta = \pm 1$ for the two non-local branches. Neglecting the Higgs mass in the external mode functions, its bulk-to-boundary propagator is

$$G_a(k; \tau) = \frac{H^2}{2k^3} (1 - iak\tau) e^{iak\tau}. \quad (4)$$

The spacetime vector propagator decomposes as

$$\begin{aligned} D_{ab,\mu\nu}(\mathbf{k}; \tau_1, \tau_2) = & \sum_{\lambda=T,L,\pm} e_{\mathbf{k},\mu}^{(\lambda)} e_{\mathbf{k},\nu}^{(\lambda)*} D_{ab}^{(\lambda)}(k; \tau_1, \tau_2) \\ & + e_{\mathbf{k},\mu}^{(T)} e_{\mathbf{k},\nu}^{(L)*} D_{ab}^{(TL)}(k; \tau_1, \tau_2) + e_{\mathbf{k},\mu}^{(L)} e_{\mathbf{k},\nu}^{(T)*} D_{ab}^{(LT)}(k; \tau_1, \tau_2), \end{aligned} \quad (5)$$

where $\mu, \nu = 0, 1, 2, 3$. In the comoving frame, the covariant polarization basis is [2]

$$e_{\mathbf{k},\mu}^{(T)} = (1, \mathbf{0}), \quad e_{\mathbf{k},\mu}^{(L)} = (0, \hat{\mathbf{k}}), \quad e_{\mathbf{k},\mu}^{(\pm)} = \frac{1}{\sqrt{2}} (0, \mathbf{e}_\theta \pm i\mathbf{e}_\phi),$$

with $(\hat{\mathbf{k}}, \mathbf{e}_\theta, \mathbf{e}_\phi)$ the orthonormal spherical basis. The label T denotes the temporal component of the longitudinal mode.

At zero chemical potential, all components can be expressed through an auxiliary scalar propagator with the same Hankel index [2, 3]:

$$D_{>}^{(s)}(k; \tau_1, \tau_2) = \frac{\pi H^2}{4} e^{-\pi \tilde{\nu}} (\tau_1 \tau_2)^{3/2} H_{i\tilde{\nu}}^{(1)}(-k\tau_1) H_{-i\tilde{\nu}}^{(2)}(-k\tau_2), \quad (6)$$

$$D_{>}^{(\pm)} = \frac{D_{>}^{(s)}}{H^2 \tau_1 \tau_2}, \quad D_{>}^{(L)} = \frac{(\partial_{\tau_1} - 2/\tau_1)(\partial_{\tau_2} - 2/\tau_2) D_{>}^{(s)}}{H^2(\tilde{\nu}^2 + 1/4)}, \quad (7)$$

$$D_{>}^{(T)} = \frac{k^2 D_{>}^{(s)}}{H^2(\tilde{\nu}^2 + 1/4)}, \quad (8)$$

$$D_{>}^{(TL)} = \frac{ik(\partial_{\tau_2} - 2/\tau_2) D_{>}^{(s)}}{H^2(\tilde{\nu}^2 + 1/4)}, \quad D_{>}^{(LT)} = -\frac{ik(\partial_{\tau_1} - 2/\tau_1) D_{>}^{(s)}}{H^2(\tilde{\nu}^2 + 1/4)}. \quad (9)$$

With spacetime indices and momentum arguments suppressed, $D_{-+} = D_{>}$, $D_{+-} = D_{<}$, and the separated-time SK propagators are

$$\begin{aligned} D_{++} &= \theta(\tau_1 - \tau_2) D_{>} + \theta(\tau_2 - \tau_1) D_{<}, \\ D_{--} &= \theta(\tau_2 - \tau_1) D_{>} + \theta(\tau_1 - \tau_2) D_{<}. \end{aligned} \quad (10)$$

for $D^{(\pm)}$, $D^{(L)}$, $D^{(TL)}$ and $D^{(LT)}$. But for D^T , the $++$ and $--$ propagators have a delta term because the temporal mode is not a physical degree of freedom:

$$\begin{aligned} D_{++}^{(T)} &= \theta(\tau_1 - \tau_2) D_{>}^{(T)} + \theta(\tau_2 - \tau_1) D_{<} + \frac{i\delta(\tau_1 - \tau_2)}{(\tilde{\nu}^2 + 1/4)a(\tau_1)^2}, \\ D_{--}^{(T)} &= \theta(\tau_2 - \tau_1) D_{>}^{(T)} + \theta(\tau_1 - \tau_2) D_{<} - \frac{i\delta(\tau_1 - \tau_2)}{(\tilde{\nu}^2 + 1/4)a(\tau_1)^2}. \end{aligned} \quad (11)$$

Then we will see that the propagators $D^{(T)}$, $D^{(TL)}$ and $D^{(LT)}$ will not contribute the leading nonlocal signal.

The leading nonlocal parts of the spatial propagators are independent of the SK indices:

$$\left[D_{\text{ab}}^{(\pm)}(k; \tau_1, \tau_2) \right]_{\text{NonL}} = \frac{(\tau_1 \tau_2)^{1/2}}{4\pi} \sum_{\eta=\pm 1} \Gamma^2(-i\eta\tilde{\nu}) \left(\frac{k^2 \tau_1 \tau_2}{4} \right)^{i\eta\tilde{\nu}}, \quad (12)$$

$$\left[D_{\text{ab}}^{(L)}(k; \tau_1, \tau_2) \right]_{\text{NonL}} = \frac{(\tau_1 \tau_2)^{1/2}}{4\pi} \sum_{\eta=\pm 1} \frac{(-1/2 + i\eta\tilde{\nu})^2}{\tilde{\nu}^2 + 1/4} \Gamma^2(-i\eta\tilde{\nu}) \left(\frac{k^2 \tau_1 \tau_2}{4} \right)^{i\eta\tilde{\nu}}. \quad (13)$$

The temporal nonlocal part is

$$\left[D_{\text{ab}}^{(T)}(k; \tau_1, \tau_2) \right]_{\text{NonL}} = \frac{k^2 (\tau_1 \tau_2)^{3/2}}{4\pi(\tilde{\nu}^2 + 1/4)} \sum_{\eta=\pm 1} \Gamma^2(-i\eta\tilde{\nu}) \left(\frac{k^2 \tau_1 \tau_2}{4} \right)^{i\eta\tilde{\nu}}. \quad (14)$$

The one-loop factorization theorem and cutting rule [4] isolate the leading nonlocal signal in the collapsed channel. We first identify the two internal lines whose removal separates the external pairs (1, 2) and (3, 4). Their momenta can be chosen as $\mathbf{q}$ and $\mathbf{K} \equiv \mathbf{q} + \mathbf{k}_s$. The relevant loop region has $q, K \sim k_s \ll k_i$. We replace these two propagators by the leading nonlocal terms of their late-time expansions, keeping their polarization tensors, and expand the remaining hard

propagators to leading order in the soft momenta. The hard propagators retain their full time dependence and SK ordering.

The cut propagators are independent of the SK indices, and their time dependence separates between the two endpoints. The time integrals thus factorize into two tree subgraphs with additional time powers supplied by the cut lines. Multiplying these subgraphs by the soft momentum integral and summing over polarizations gives the nonlocal signal. The branches $\eta = \pm 1$ combine into the oscillatory signal retained below. Higher terms in the soft expansion give subleading corrections in the collapsed limit.

The propagators $D^{(T)}$, $D^{(TL)}$, and $D^{(LT)}$ do not contribute to the leading nonlocal signal. On a soft line, each temporal mode function $B^{(T)}$ carries an additional factor of $k\tau$ relative to the spatial modes $B^{(L)}$ and $B^{(\pm)}$. Their nonanalytic contributions are therefore suppressed in the collapsed limit. The contact term in $D^{(T)}$ is analytic in the soft momentum and does not contribute to the nonlocal signal. On a hard line, a temporal endpoint has a vanishing polarization contraction with the spatial components of the attached soft line. A nonzero contraction would require a temporal soft component, which is already suppressed. We therefore omit propagators with temporal components when computing the leading signal below.

All required soft momentum integrals reduce to five dimensionless functions f_r , $r = 0, 1, 2, 3, 4$. With $z \equiv \hat{\mathbf{q}} \cdot \hat{\mathbf{k}}_s$, we define

$$\int \frac{d^3\mathbf{q}}{(2\pi)^3} q^{-2\alpha} K^{-2\beta} z^r = \frac{k_s^{3-2\alpha-2\beta}}{(4\pi)^{3/2}} f_r(\alpha, \beta), \quad r = 0, 1, 2, 3, 4. \quad (15)$$

These expressions are understood by analytic continuation from the convergent domain. The first three functions are

$$f_0(\alpha, \beta) = \frac{\Gamma(\alpha + \beta - 3/2)\Gamma(3/2 - \alpha)\Gamma(3/2 - \beta)}{\Gamma(3 - \alpha - \beta)\Gamma(\alpha)\Gamma(\beta)}, \quad (16)$$

$$f_1(\alpha, \beta) = \frac{1}{2} \left[f_0(\alpha + \frac{1}{2}, \beta - 1) - f_0(\alpha - \frac{1}{2}, \beta) - f_0(\alpha + \frac{1}{2}, \beta) \right], \quad (17)$$

$$f_2(\alpha, \beta) = \frac{1}{4} \left[f_0(\alpha + 1, \beta - 2) + f_0(\alpha - 1, \beta) + f_0(\alpha + 1, \beta) - 2f_0(\alpha, \beta - 1) - 2f_0(\alpha + 1, \beta - 1) + 2f_0(\alpha, \beta) \right]. \quad (18)$$

The functions f_3 and f_4 are likewise finite sums of f_0 ; their explicit expressions are collected in App. A.

For the nonlocal branch η , it is convenient to set $x = -i\eta\tilde{\nu}$ and define

$$\mathcal{J}_r(m, n; x) \equiv f_r \left(x - \frac{m}{2}, x + \frac{n}{2} \right), \quad r = 0, 1, 2, 3, 4. \quad (19)$$

Here m, n are integer shifts of momentum powers, unrelated to m_V . In particular,

$$\int \frac{d^3\mathbf{q}}{(2\pi)^3} \left(\frac{q}{2} \right)^{-2x} \left(\frac{K}{2} \right)^{-2x} q^m K^{-n} z^r = \frac{4^{2x} k_s^{3-4x+m-n}}{(4\pi)^{3/2}} \mathcal{J}_r(m, n; x). \quad (20)$$

2.2 Bubble Diagram

The bubble contains two quartic vertices, each attached to a pair of external Higgs modes. With the four external momenta understood, the s -channel contribution is

$$\begin{aligned} \langle \delta h^4 \rangle'_{\text{Bub}} &= \frac{g^4}{8} \sum_{\mathbf{a}, \mathbf{b}=\pm 1} (-\mathbf{ab}) \int_{-\infty}^0 \frac{d\tau_1}{|H\tau_1|^2} \int_{-\infty}^0 \frac{d\tau_2}{|H\tau_2|^2} G_{\mathbf{a}}(k_1; \tau_1) G_{\mathbf{a}}(k_2; \tau_1) G_{\mathbf{b}}(k_3; \tau_2) G_{\mathbf{b}}(k_4; \tau_2) \\ &\quad \times \int \frac{d^3 \mathbf{q}}{(2\pi)^3} D_{\mathbf{ab}, ij}(\mathbf{q}; \tau_1, \tau_2) D_{\mathbf{ba}, ji}(\mathbf{K}; \tau_2, \tau_1), \quad \mathbf{K} = \mathbf{q} + \mathbf{k}_s. \end{aligned} \quad (21)$$

The symmetry factor $1/2$ is included. Throughout this chapter, the $W^\pm$ contribution is obtained by multiplying the expressions by 2, and the Z contribution by replacing $g \rightarrow g/c_W$, with the corresponding mass used in each case.

Replacing the two soft propagators by (12)–(13) removes their SK-index dependence. After summing the polarizations, the oscillatory part of the loop integral is

$$\begin{aligned} &\left[\int \frac{d^3 \mathbf{q}}{(2\pi)^3} D_{\mathbf{ab}, ij}(\mathbf{q}; \tau_1, \tau_2) D_{\mathbf{ba}, ji}(\mathbf{K}; \tau_2, \tau_1) \right]_{\text{NonL}} \\ &\simeq \sum_{\eta=\pm 1} (\tau_1 \tau_2)^{1+2i\eta\tilde{\nu}} \frac{\Gamma^4(-i\eta\tilde{\nu})}{(4\pi)^{7/2}} 4^{-2i\eta\tilde{\nu}} k_s^{3+4i\eta\tilde{\nu}} \mathcal{I}_{\text{Bub}}(-i\eta\tilde{\nu}). \end{aligned} \quad (22)$$

The dimensionless polarization sum has the compact form

$$\mathcal{I}_{\text{Bub}}(x) = \frac{3+2x}{1-2x} \mathcal{J}_0(0, 0; x) + \frac{16x^2}{(1-2x)^2} [\mathcal{J}_0(2, 2; x) + 2\mathcal{J}_1(1, 2; x) + \mathcal{J}_2(0, 2; x)], \quad (23)$$

with $x = -i\eta\tilde{\nu}$ as in (19). The separate polarization contributions are given in App. B.

The remaining time integrals factorize. For the left subgraph, at $k_1 \simeq k_2 \simeq k_{12}/2$, we find

$$\sum_{\mathbf{a}=\pm 1} i\mathbf{a} \int_{-\infty}^0 \frac{d\tau}{|H\tau|^2} G_{\mathbf{a}}(k_1; \tau) G_{\mathbf{a}}(k_2; \tau) (-\tau)^{1+2i\eta\tilde{\nu}} \simeq \frac{H^2 k_{12}^{-2i\eta\tilde{\nu}}}{4k_1^3 k_2^3} \mathcal{P}_\eta^{(1)}(\tilde{\nu}), \quad (24)$$

$$\mathcal{P}_\eta^{(1)}(\tilde{\nu}) = \frac{2+i\eta\tilde{\nu}}{2i\eta\tilde{\nu}} i \sinh(\pi\eta\tilde{\nu}) \Gamma(2+2i\eta\tilde{\nu}). \quad (25)$$

The right vertex gives the same expression with $(k_1, k_2, k_{12}) \rightarrow (k_3, k_4, k_{34})$. Combining the two vertices with (22), we obtain

$$[\langle \delta h^4 \rangle'_{\text{Bub}}]_{\text{NonL}} \simeq \frac{g^4 H^4 k_s^3}{128 k_1^3 k_2^3 k_3^3 k_4^3} \left[\left(\frac{k_s^2}{4k_{12}k_{34}} \right)^{2i\tilde{\nu}} C^{\text{Bub}}(\tilde{\nu}) + \text{c.c.} \right], \quad (26)$$

where the complex signal coefficient is

$$C^{\text{Bub}}(\tilde{\nu}) = \frac{\Gamma^4(-i\tilde{\nu})}{(4\pi)^{7/2}} [\mathcal{P}_+^{(1)}(\tilde{\nu})]^2 \mathcal{I}_{\text{Bub}}(-i\tilde{\nu}). \quad (27)$$

This definition separates the loop and time integrals from the couplings and diagram symmetry factor. The bracket in (26) equals

$$2|C^{\text{Bub}}(\tilde{\nu})| \cos \left[2\tilde{\nu} \log \left(\frac{k_s^2}{4k_{12}k_{34}} \right) + \arg C^{\text{Bub}}(\tilde{\nu}) \right]. \quad (28)$$

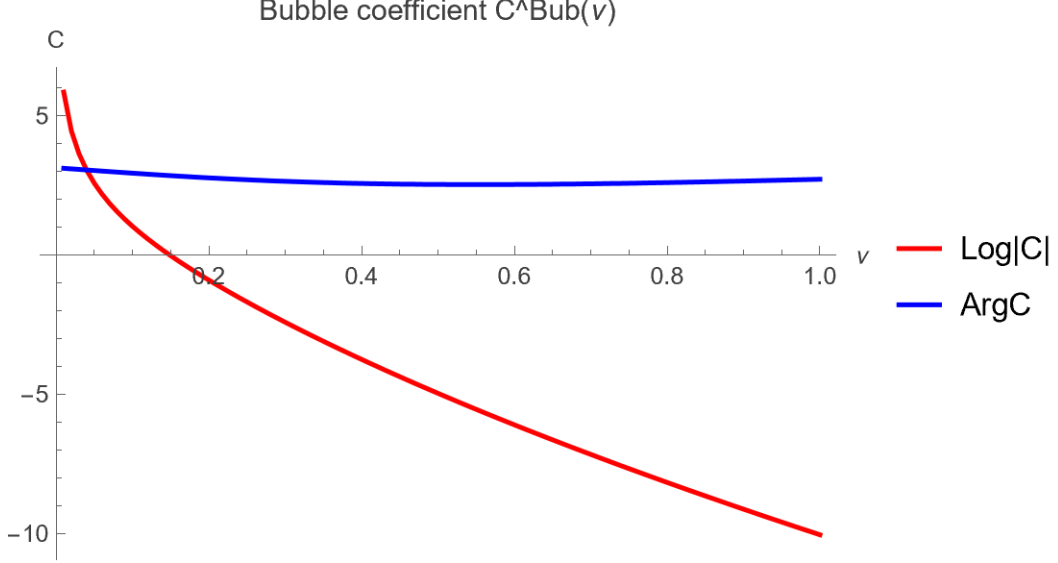


Figure 1: The bubble coefficient defined in (27). The red and blue curves show $\log |C^{Bub}|$ and $\arg C^{Bub}$ in radians, respectively, as functions of $\tilde{\nu}$.

2.3 Triangle Diagram

The triangle contains two cubic vertices and one quartic vertex. We first place the external modes $\mathbf{k}_1, \mathbf{k}_2$ at the cubic vertices and $\mathbf{k}_3, \mathbf{k}_4$ at the quartic vertex. The cubic interaction for one real vector component is $a^2 g^2 h_0 \delta h V_\mu V^\mu / 4$, with vertex coefficient $g^2 h_0 / 2$. Together with the quartic coefficient $g^2 / 2$, this gives an overall factor $g^6 h_0^2 / 8$; the triangle has unit diagram symmetry factor. In the conventions of Sec. 2.2, its SK expression is

$$\begin{aligned} \langle \delta h^4 \rangle'_{\text{Tri1}} &= \frac{g^6 h_0^2}{8} \sum_{\mathbf{a}, \mathbf{b}, \mathbf{c} = \pm 1} (-iabc) \int_{-\infty}^0 \prod_{v=1}^3 \frac{d\tau_v}{|H\tau_v|^2} G_{\mathbf{a}}(k_1; \tau_1) G_{\mathbf{b}}(k_2; \tau_2) G_{\mathbf{c}}(k_3; \tau_3) G_{\mathbf{c}}(k_4; \tau_3) \\ &\quad \times \int \frac{d^3 \mathbf{q}}{(2\pi)^3} D_{\mathbf{ab}, ij}(\mathbf{q} + \mathbf{k}_1; \tau_1, \tau_2) D_{\mathbf{bc}, j\ell}(\mathbf{K}; \tau_2, \tau_3) D_{\mathbf{ca}, \ell i}(\mathbf{q}; \tau_3, \tau_1). \end{aligned} \quad (29)$$

The nonlocal signal comes from $q, K \sim k_s \ll k_1$. We cut the $\mathbf{q}$ and $\mathbf{K}$ lines and evaluate the remaining hard propagator at $\mathbf{q} + \mathbf{k}_1 \simeq \mathbf{k}_1$. The latter retains its full time dependence and SK ordering.

After summing the soft polarizations, the cut loop takes the form

$$\begin{aligned} &\left[\int \frac{d^3 \mathbf{q}}{(2\pi)^3} D_{\mathbf{ab}, ij}(\mathbf{q} + \mathbf{k}_1; \tau_1, \tau_2) D_{\mathbf{bc}, j\ell}(\mathbf{K}; \tau_2, \tau_3) D_{\mathbf{ca}, \ell i}(\mathbf{q}; \tau_3, \tau_1) \right]_{\text{NonL}} \\ &\simeq \sum_{\lambda=L, \pm} \sum_{\eta=\pm 1} D_{\mathbf{ab}}^{(\lambda)}(k_1; \tau_1, \tau_2) (\tau_1 \tau_2)^{1/2 + i\eta\tilde{\nu}} (-\tau_3)^{1 + 2i\eta\tilde{\nu}} \frac{\Gamma^4(-i\eta\tilde{\nu})}{(4\pi)^{7/2}} 4^{-2i\eta\tilde{\nu}} k_s^{3 + 4i\eta\tilde{\nu}} \mathcal{I}_{\text{Tri}}^\lambda(-i\eta\tilde{\nu}, \theta_1). \end{aligned} \quad (30)$$

This equation defines the dimensionless coefficients $\mathcal{I}_{\text{Tri}}^\lambda(x, \theta)$, with $x = -i\eta\tilde{\nu}$. Rotational symmetry and polarization completeness imply

$$\mathcal{I}_{\text{Tri}}^L(x, \theta) = \sin^2 \theta \mathcal{I}_{\text{Tri}}^L(x, \pi/2) + \cos^2 \theta \mathcal{I}_{\text{Tri}}^L(x, 0), \quad (31)$$

$$\mathcal{I}_{\text{Tri}}^+(x, \theta) = \mathcal{I}_{\text{Tri}}^-(x, \theta) = \frac{1}{2} [\mathcal{I}_{\text{Bub}}(x) - \mathcal{I}_{\text{Tri}}^L(x, \theta)]. \quad (32)$$

Thus only one longitudinal coefficient is needed in addition to the bubble sum. Its expression in terms of $\mathcal{J}_r$ is given in App. C. After simplification, all three coefficients equal $\mathcal{I}_{\text{Bub}}(x)/3$; the apparent angular dependence cancels, as shown in (53).

The quartic vertex gives the time integral $\mathcal{P}_\eta^{(1)}$ in (25). The two cubic vertices and the hard line instead define

$$\begin{aligned} \sum_{a,b=\pm 1} (-ab) \int_{-\infty}^0 \frac{d\tau_1 d\tau_2}{|H\tau_1|^2 |H\tau_2|^2} G_a(k_1; \tau_1) G_b(k_2; \tau_2) D_{ab}^{(\lambda)}(k_1; \tau_1, \tau_2) (\tau_1 \tau_2)^{1/2+i\eta\tilde{\nu}} \\ \simeq \frac{k_{12}^{-2i\eta\tilde{\nu}}}{4k_1^3 k_2^3} \mathcal{P}_\eta^{(2),\lambda}(\tilde{\nu}). \end{aligned} \quad (33)$$

At leading order, $k_1 \simeq k_2 \simeq k_{12}/2$. The energy entering each end of the hard propagator then equals its momentum magnitude. The integral therefore reduces to the double-folded scalar seeds of Ref. [3], which we denote by $\mathcal{M}_{ab}^{p_1 p_2} \equiv \tilde{\mathcal{I}}_{ab}^{p_1 p_2}(1, 1)$. For the transverse modes this follows directly from (7). For the longitudinal mode, the differential relation also produces a contact contribution [2],

$$\mathcal{P}_\eta^{(2),L}(\tilde{\nu}) \Big|_{\text{contact}} = \frac{\mathcal{P}_\eta^{(1)}(\tilde{\nu})}{\tilde{\nu}^2 + 1/4}. \quad (34)$$

We retain this term together with the scalar-seed contributions. The required seed exponents include $p_0 = -5/2 + i\eta\tilde{\nu}$, on the boundary of the convergence domain. We define the result by analytic continuation from $p_0 + \epsilon$, $\epsilon > 0$, and take the finite part of the assembled expression at $\epsilon = 0$. The seed normalization, closed forms, and combinations giving $\mathcal{P}_\eta^{(2),\lambda}$ are collected in App. C.

The leading signal of the first triangle is

$$\left[\langle \delta h^4 \rangle'_{\text{Tri1}} \right]_{\text{NonL}} \simeq \frac{g^6 h_0^2 H^2 k_s^3}{128 k_1^3 k_2^3 k_3^3 k_4^3} \left[\left(\frac{k_s^2}{4k_{12} k_{34}} \right)^{2i\tilde{\nu}} C^{\text{Tri}}(\tilde{\nu}, \theta_1) + \text{c.c.} \right], \quad (35)$$

where

$$C^{\text{Tri}}(\tilde{\nu}, \theta) = \frac{\Gamma^4(-i\tilde{\nu})}{(4\pi)^{7/2}} \mathcal{P}_+^{(1)}(\tilde{\nu}) \sum_{\lambda=L,\pm} \mathcal{P}_+^{(2),\lambda}(\tilde{\nu}) \mathcal{I}_{\text{Tri}}^\lambda(-i\tilde{\nu}, \theta). \quad (36)$$

Since $\mathcal{P}_\eta^{(2),+} = \mathcal{P}_\eta^{(2),-}$, the two transverse helicities contribute equally. The simplified loop coefficients make C^{Tri} independent of θ at this order; its reduced expression is given in (54).

The second triangle has the quartic vertex on the (1, 2) side and follows by (1, 2) $\leftrightarrow$ (3, 4). Adding the two placements gives

$$\left[\langle \delta h^4 \rangle'_{\text{Tri}} \right]_{\text{NonL}} \simeq \frac{g^6 h_0^2 H^2 k_s^3}{128 k_1^3 k_2^3 k_3^3 k_4^3} \left\{ \left(\frac{k_s^2}{4k_{12} k_{34}} \right)^{2i\tilde{\nu}} \left[C^{\text{Tri}}(\tilde{\nu}, \theta_1) + C^{\text{Tri}}(\tilde{\nu}, \theta_3) \right] + \text{c.c.} \right\}. \quad (37)$$

The nonanalytic momentum dependence is the same as in the bubble, while the hard propagator changes the amplitude and phase. Figure 2 shows the coefficient of a single triangle at $\theta_1 = 0$, with its Boltzmann suppression included. Both curves are independent of the angle between the hard momentum and $\mathbf{k}_s$ at leading order in the collapsed limit; $\theta_1 = 0$ is only a plotting convention. Couplings remain outside C^{Tri} as in (35).

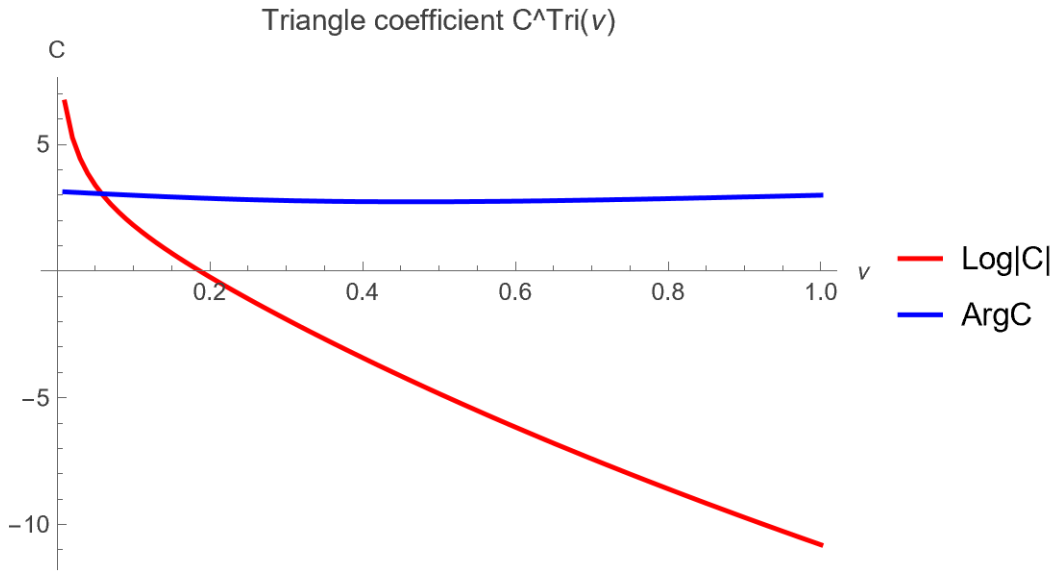


Figure 2: The coefficient $C^{\text{Tri}}(\tilde{\nu}, 0)$ of a single triangle, defined in (36). The red and blue curves show $\log |C^{\text{Tri}}|$ and $\arg C^{\text{Tri}}$ in radians, respectively. Both curves are independent of the momentum angle at leading order in the collapsed limit; the choice $\theta_1 = 0$ is arbitrary. The horizontal variable is $\tilde{\nu}$ and the Boltzmann suppression is included. The sum of the two triangle placements is given separately in (37).

2.4 Box Diagram

We consider the box with external legs in cyclic order $(1, 2, 3, 4)$. The four cubic vertices give the coupling factor $(g^2 h_0/2)^4 = g^8 h_0^4/16$, and the diagram symmetry factor is unity. All results in this subsection refer to this single ordering.

Choose the loop momenta along the four successive internal lines as $\mathbf{q} + \mathbf{k}_1$, $\mathbf{K} = \mathbf{q} + \mathbf{k}_s$, $\mathbf{q} - \mathbf{k}_4$, and $\mathbf{q}$. In the collapsed limit, the leading nonlocal signal comes from the region $q, K \sim k_s$. Cutting these two soft lines leaves two hard propagators, with momenta approaching $\mathbf{k}_1$ and $\mathbf{k}_3$. Their polarizations are denoted by $\lambda_1, \lambda_3 = L, \pm$. The soft loop gives the dimensionless coefficient $\mathcal{I}_{\text{Box}}^{\lambda_1 \lambda_3}(-i\eta\tilde{\nu}, \theta_1, \theta_3, \phi_3)$ defined in (60). Its common momentum factor is $4^{-2i\eta\tilde{\nu}} k_s^{3+4i\eta\tilde{\nu}}$, as for the bubble and triangle.

The time integrals factorize into two copies of (33), one for each hard line. They contribute $\mathcal{P}_\eta^{(2), \lambda_1}(\tilde{\nu})$ and $\mathcal{P}_\eta^{(2), \lambda_3}(\tilde{\nu})$, including the longitudinal contact terms. Thus no additional time-integral seed is required. Combining the two branches gives

$$\begin{aligned} \left[\langle \delta h^4 \rangle'_{\text{Box}(1234)} \right]_{\text{NonL}} &\simeq \frac{g^8 h_0^4 k_s^3}{256 k_1^3 k_2^3 k_3^3 k_4^3} \left[\left(\frac{k_s^2}{4k_{12}k_{34}} \right)^{2i\tilde{\nu}} \right. \\ &\quad \left. \times C^{\text{Box}}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) + \text{c.c.} \right], \end{aligned} \quad (38)$$

where

$$\begin{aligned} C^{\text{Box}}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) &= \frac{\Gamma^4(-i\tilde{\nu})}{(4\pi)^{7/2}} \sum_{\lambda_1, \lambda_3 = L, \pm} \mathcal{P}_+^{(2), \lambda_1}(\tilde{\nu}) \mathcal{P}_+^{(2), \lambda_3}(\tilde{\nu}) \\ &\quad \times \mathcal{I}_{\text{Box}}^{\lambda_1 \lambda_3}(-i\tilde{\nu}, \theta_1, \theta_3, \phi_3). \end{aligned} \quad (39)$$

The normalization follows the vertex convention of Secs. 2.2 and 2.3. The coefficient C^{Box} is independent of the coupling g .

Since the hard transverse propagators have equal time integrals, the polarization sum reduces to the LL , LH , and HH groups in (61). Their explicit expressions are given in App. D. In particular, (65) shows that all angular dependence is carried by $\mathcal{I}_{\text{Box}}^{LL}$ and multiplied by $[\mathcal{P}_+^{(2), L} - \mathcal{P}_+^{(2), +}]^2$. The difference between the longitudinal and transverse hard-line time integrals therefore controls the angular dependence of the box signal.

The result contains five angular structures:

$$\begin{aligned} C^{\text{Box}}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) &= A_1(\tilde{\nu}) + A_2(\tilde{\nu})(\cos^2 \theta_1 + \cos^2 \theta_3) + A_3(\tilde{\nu}) \cos^2 \theta_1 \cos^2 \theta_3 \\ &\quad + B(\tilde{\nu}) \sin(2\theta_1) \sin(2\theta_3) \cos \phi_3 \\ &\quad + D(\tilde{\nu}) \sin^2 \theta_1 \sin^2 \theta_3 \cos(2\phi_3). \end{aligned} \quad (40)$$

The five complex functions depend only on $\tilde{\nu}$; their angular projections are given in App. D.2. The signal is symmetric under $\theta_1 \leftrightarrow \theta_3$ and $\phi_3 \rightarrow -\phi_3$. Its oscillatory momentum dependence is the same as in (28), with an angle-dependent amplitude and phase supplied by C^{Box} .

Figures 3 and 4 show the five coefficient functions, including their Boltzmann suppression. Their magnitudes decrease over the displayed mass range. The jumps in the plotted phases reflect the principal branch of $\arg$, defined modulo 2π .

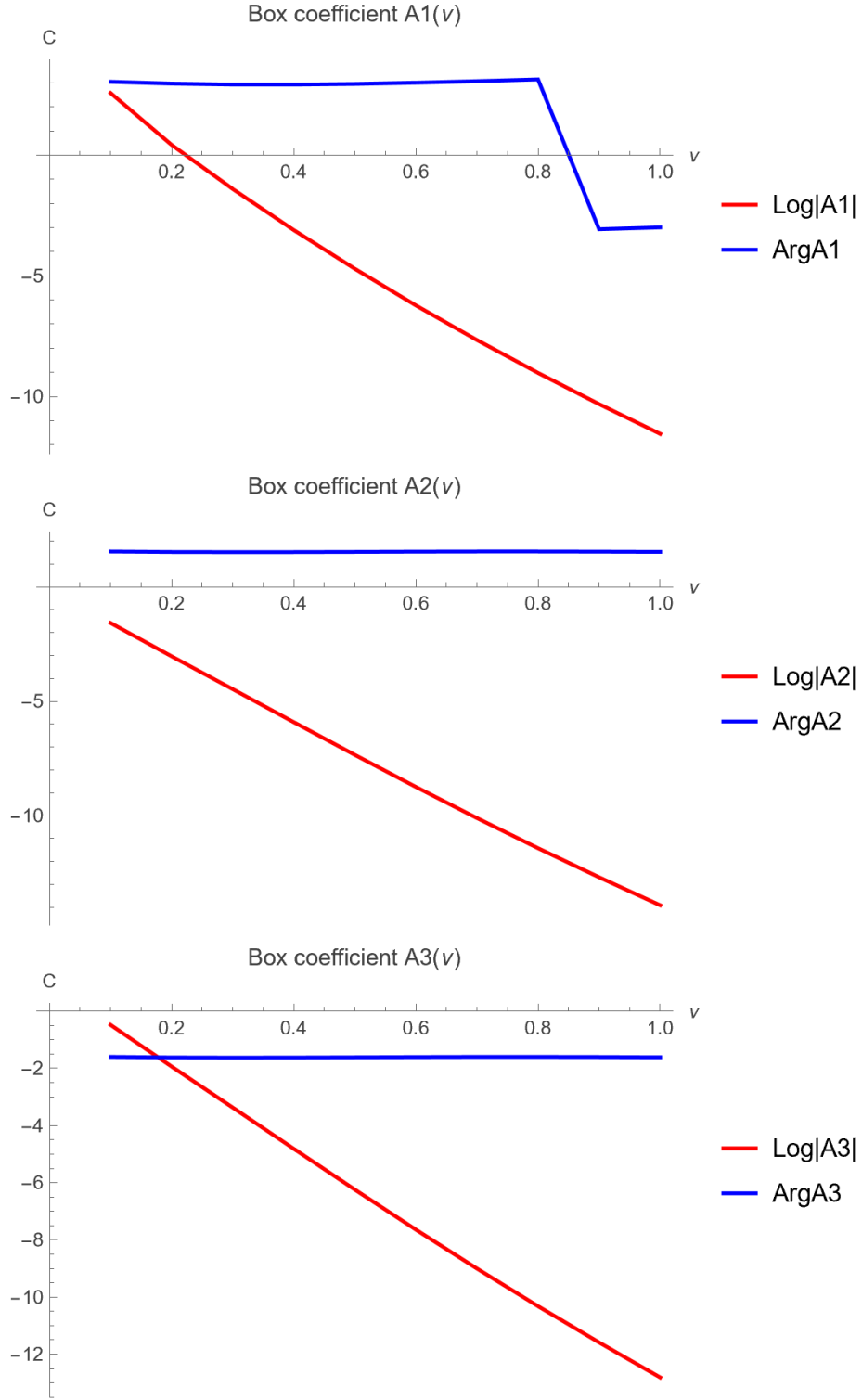


Figure 3: The angular coefficients A_1 , A_2 , and A_3 (top to bottom) in (40) for the single box ordering (1, 2, 3, 4). Red curves show the logarithmic magnitudes and blue curves show the principal phases in radians. The horizontal variable is $\tilde{\nu}$; Boltzmann suppression is included.

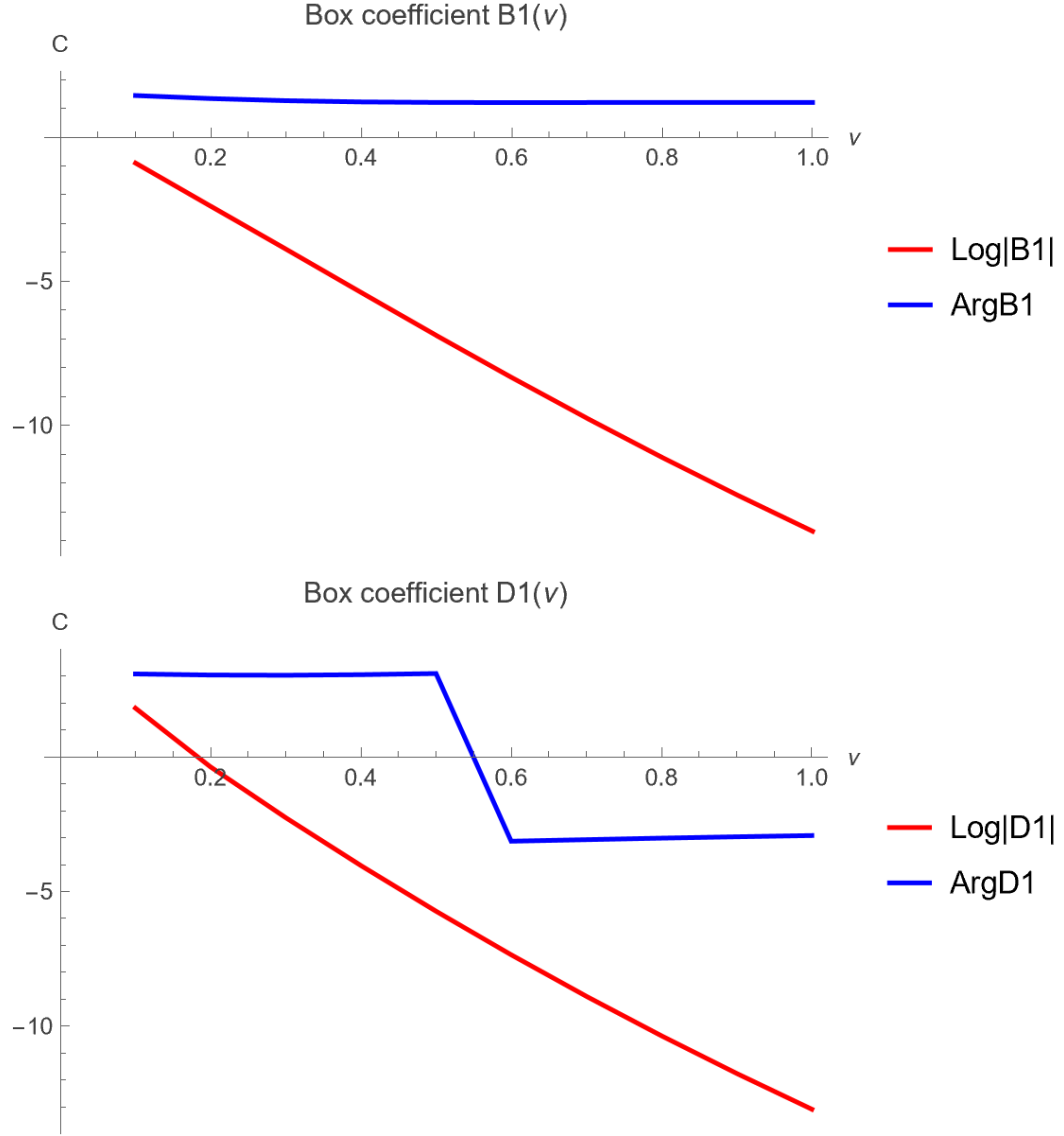


Figure 4: The angular coefficients B (top, labelled $B1$) and D (bottom, labelled $D1$) in (40). The plotting conventions and single-ordering normalization are the same as in Fig. 3, with Boltzmann suppression included.

3 Top Quark Signal

4 Conclusions

A Loop Integral Formulae

The identity $z = (K^2 - q^2 - k_s^2)/(2qk_s)$ reduces all angular moments in (15) to the scalar integral f_0 :

$$f_r(\alpha, \beta) = \frac{1}{2^r} \sum_{u+v+w=r} \frac{r!}{u!v!w!} (-1)^{v+w} f_0\left(\alpha + \frac{r}{2} - v, \beta - u\right), \quad (41)$$

where the sum runs over nonnegative integers. For the third and fourth moments this gives

$$\begin{aligned} f_3(\alpha, \beta) = \frac{1}{8} & \left[f_0(\alpha + \tfrac{3}{2}, \beta - 3) - 3f_0(\alpha + \tfrac{3}{2}, \beta - 2) - 3f_0(\alpha + \tfrac{1}{2}, \beta - 2) \right. \\ & + 3f_0(\alpha + \tfrac{3}{2}, \beta - 1) + 6f_0(\alpha + \tfrac{1}{2}, \beta - 1) + 3f_0(\alpha - \tfrac{1}{2}, \beta - 1) \\ & \left. - f_0(\alpha + \tfrac{3}{2}, \beta) - 3f_0(\alpha + \tfrac{1}{2}, \beta) - 3f_0(\alpha - \tfrac{1}{2}, \beta) - f_0(\alpha - \tfrac{3}{2}, \beta) \right], \end{aligned} \quad (42)$$

$$\begin{aligned} f_4(\alpha, \beta) = \frac{1}{16} & \left[f_0(\alpha + 2, \beta - 4) - 4f_0(\alpha + 2, \beta - 3) - 4f_0(\alpha + 1, \beta - 3) \right. \\ & + 6f_0(\alpha + 2, \beta - 2) + 12f_0(\alpha + 1, \beta - 2) + 6f_0(\alpha, \beta - 2) \\ & - 4f_0(\alpha + 2, \beta - 1) - 12f_0(\alpha + 1, \beta - 1) - 12f_0(\alpha, \beta - 1) \\ & - 4f_0(\alpha - 1, \beta - 1) + f_0(\alpha + 2, \beta) + 4f_0(\alpha + 1, \beta) \\ & \left. + 6f_0(\alpha, \beta) + 4f_0(\alpha - 1, \beta) + f_0(\alpha - 2, \beta) \right]. \end{aligned} \quad (43)$$

Thus $f_1, \dots, f_4$ introduce no special functions beyond the Gamma functions in (16).

B Bubble Polarization Coefficients

For $x = -i\eta\tilde{\nu}$, the relative weight of a longitudinal nonlocal propagator is

$$\beta(x) \equiv \frac{(x + 1/2)^2}{1/4 - x^2} = \frac{1 + 2x}{1 - 2x}. \quad (44)$$

Define $\mathcal{I}_{\sigma_1\sigma_2}(x)$ by restricting the loop in (22) to the polarizations (σ_1, σ_2) and replacing $\mathcal{I}_{Bub}$ on its right-hand side by $\mathcal{I}_{\sigma_1\sigma_2}$. The independent coefficients are

$$\mathcal{I}_{LL}(x) = [\beta(x)]^2 [\mathcal{J}_0(2, 2; x) + 2\mathcal{J}_1(1, 2; x) + \mathcal{J}_2(0, 2; x)], \quad (45)$$

$$\mathcal{I}_{L+}(x) = \frac{\beta(x)}{2} [\mathcal{J}_0(0, 2; x) - \mathcal{J}_2(0, 2; x)], \quad (46)$$

$$\begin{aligned} \mathcal{I}_{++}(x) = \frac{1}{4} & [\mathcal{J}_0(0, 0; x) + \mathcal{J}_0(2, 2; x) + \mathcal{J}_2(0, 2; x) \\ & + 2\mathcal{J}_0(1, 1; x) + 2\mathcal{J}_1(1, 2; x) + 2\mathcal{J}_1(0, 1; x)], \end{aligned} \quad (47)$$

$$\begin{aligned} \mathcal{I}_{+-}(x) = \frac{1}{4} & [\mathcal{J}_0(0, 0; x) + \mathcal{J}_0(2, 2; x) + \mathcal{J}_2(0, 2; x) \\ & - 2\mathcal{J}_0(1, 1; x) + 2\mathcal{J}_1(1, 2; x) - 2\mathcal{J}_1(0, 1; x)]. \end{aligned} \quad (48)$$

The remaining entries satisfy $\mathcal{I}_{--} = \mathcal{I}_{++}$, $\mathcal{I}_{-+} = \mathcal{I}_{+-}$, and $\mathcal{I}_{L-} = \mathcal{I}_{+L} = \mathcal{I}_{-L} = \mathcal{I}_{L+}$. Consequently,

$$\mathcal{I}_{Bub} = \mathcal{I}_{LL} + 4\mathcal{I}_{L+} + 2\mathcal{I}_{++} + 2\mathcal{I}_{+-}, \quad (49)$$

which reduces to (23) using the moment relations in App. A.

For the $\eta = +1$ branch, the fully reduced sum is

$$\mathcal{I}_{Bub}(-i\tilde{\nu}) = \frac{3 \cdot 2^{-4-4i\tilde{\nu}} (3i - 2\tilde{\nu})^2 \Gamma(-3/2 - 2i\tilde{\nu}) \Gamma(3 + 2i\tilde{\nu}) \sinh^2(\pi\tilde{\nu})}{\pi[1 + (3i - 2\tilde{\nu})\tilde{\nu}]^2}. \quad (50)$$

Substituting this into (27) gives

$$\begin{aligned} C^{Bub}(\tilde{\nu}) &= \frac{3(\tilde{\nu} - i)(\tilde{\nu} - 2i)^4}{256\pi^{7/2}\tilde{\nu}^2(2\tilde{\nu} - i)} \Gamma^2(-2 - i\tilde{\nu}) \Gamma^2(5/2 + i\tilde{\nu}) \\ &\quad \times \Gamma(-3/2 - 2i\tilde{\nu}) \Gamma(1 + 2i\tilde{\nu}) \sinh^2(\pi\tilde{\nu}). \end{aligned} \quad (51)$$

These expressions retain the Boltzmann suppression. For real $\tilde{\nu} > 0$, the opposite nonlocal branch is their complex conjugate.

C Triangle Coefficients

C.1 Polarization sum

Let $x = -i\eta\tilde{\nu}$ and let $\beta(x)$ be the longitudinal weight in (44). Summing the nine soft-polarization combinations gives

$$\begin{aligned} \mathcal{I}_{Tri}^L(x, \theta) &= [\beta(x)]^2 \left\{ \frac{\sin^2 \theta}{2} [\mathcal{J}_0(2, 2; x) + \mathcal{J}_1(1, 2; x) - \mathcal{J}_2(2, 2; x) - \mathcal{J}_3(1, 2; x)] \right. \\ &\quad \left. + \cos^2 \theta [\mathcal{J}_2(2, 2; x) + \mathcal{J}_3(1, 2; x) + \mathcal{J}_1(1, 2; x) + \mathcal{J}_2(0, 2; x)] \right\} \\ &\quad + \frac{\beta(x)(1 + \cos^2 \theta)}{2} [\mathcal{J}_0(0, 2; x) - \mathcal{J}_2(0, 2; x)] \\ &\quad + \frac{\sin^2 \theta}{2} [\mathcal{J}_2(2, 2; x) + \mathcal{J}_3(1, 2; x) + \mathcal{J}_1(1, 2; x) + \mathcal{J}_2(0, 2; x) + \mathcal{J}_0(0, 0; x)] \\ &\quad + \cos^2 \theta [\mathcal{J}_0(2, 2; x) - \mathcal{J}_2(2, 2; x) + \mathcal{J}_1(1, 2; x) - \mathcal{J}_3(1, 2; x)]. \end{aligned} \quad (52)$$

The terms proportional to β^2 , β , and 1 arise from two longitudinal, mixed, and two transverse soft lines, respectively. The hard transverse coefficients follow from (32); each refers to one helicity.

Reducing the Gamma functions in the moment integrals yields

$$\begin{aligned} \mathcal{I}_{Tri}^L(-i\tilde{\nu}, \theta) &= \mathcal{I}_{Tri}^+(-i\tilde{\nu}, \theta) = \mathcal{I}_{Tri}^-(-i\tilde{\nu}, \theta) = \frac{1}{3} \mathcal{I}_{Bub}(-i\tilde{\nu}) \\ &= \frac{2^{-4-4i\tilde{\nu}} (3i - 2\tilde{\nu})^2 \Gamma(-3/2 - 2i\tilde{\nu}) \Gamma(3 + 2i\tilde{\nu}) \sinh^2(\pi\tilde{\nu})}{\pi[1 + (3i - 2\tilde{\nu})\tilde{\nu}]^2}. \end{aligned} \quad (53)$$

Thus the angular dependence of the separate soft-polarization contributions cancels in their sum. The coefficient of a single triangle reduces to

$$C^{Tri}(\tilde{\nu}, \theta) = \frac{\Gamma^4(-i\tilde{\nu})}{3(4\pi)^{7/2}} \mathcal{P}_+^{(1)}(\tilde{\nu}) \left[\mathcal{P}_+^{(2),L}(\tilde{\nu}) + 2\mathcal{P}_+^{(2),+}(\tilde{\nu}) \right] \mathcal{I}_{Bub}(-i\tilde{\nu}). \quad (54)$$

C.2 Time-integral coefficients

For the auxiliary scalar propagator of (6), the double-folded seed is defined by

$$\mathcal{M}_{ab}^{p_1 p_2} \equiv (-ab) k^{5+p_{12}} \int_{-\infty}^0 d\tau_1 d\tau_2 (-\tau_1)^{p_1} (-\tau_2)^{p_2} e^{ia k \tau_1 + ib k \tau_2} \frac{D_{ab}^{(s)}(k; \tau_1, \tau_2)}{H^2}, \quad (55)$$

where $p_{12} = p_1 + p_2$ and $\bar{p}_{12} = p_1 - p_2$. This is the seed $\tilde{\mathcal{I}}_{ab}^{p_1 p_2}(1, 1)$ of Ref. [3], with H restored explicitly. All seeds below have mass index $\tilde{\nu}$. The closed expressions are

$$\mathcal{M}_{\pm\mp}^{p_1 p_2} = \frac{e^{\mp i\pi \bar{p}_{12}/2}}{2^{5+p_{12}}} \prod_{j=1}^2 \frac{\Gamma(5/2 + p_j - i\tilde{\nu}) \Gamma(5/2 + p_j + i\tilde{\nu})}{\Gamma(3 + p_j)}, \quad (56)$$

$$\begin{aligned} \mathcal{M}_{\pm\pm}^{p_1 p_2} &= \frac{\pm i e^{\mp i\pi p_{12}/2} e^{-\pi\tilde{\nu}}}{2^{5+p_{12}}} \prod_{j=1}^2 \frac{\Gamma(5/2 + p_j - i\tilde{\nu}) \Gamma(5/2 + p_j + i\tilde{\nu})}{\Gamma(3 + p_j)} \\ &\quad - \frac{e^{\mp i\pi p_{12}/2}}{2^{5+p_{12}}} \Gamma(5 + p_{12}) \Gamma(5/2 + p_1 \pm i\tilde{\nu}) \Gamma(5/2 + p_2 \pm i\tilde{\nu}) \\ &\quad \times {}_3\tilde{F}_2 \left[\begin{matrix} 5 + p_{12}, 1/2 \pm i\tilde{\nu}, 1 \\ 7/2 + p_1 \pm i\tilde{\nu}, 7/2 + p_2 \pm i\tilde{\nu} \end{matrix} \middle| 1 \right]. \end{aligned} \quad (57)$$

Here ${}_3\tilde{F}_2(a, b, c; d, e; 1) = {}_3F_2(a, b, c; d, e; 1)/[\Gamma(d)\Gamma(e)]$ is the regularized hypergeometric function.

Set $p_\epsilon = -5/2 + \epsilon + i\eta\tilde{\nu}$. The two transverse coefficients are

$$\mathcal{P}_\eta^{(2),\pm}(\tilde{\nu}) = 2^{2i\eta\tilde{\nu}} \text{FP}_{\epsilon=0} \sum_{a,b=\pm 1} \left[\mathcal{M}_{ab}^{p_\epsilon, p_\epsilon} + ia \mathcal{M}_{ab}^{p_\epsilon+1, p_\epsilon} + ib \mathcal{M}_{ab}^{p_\epsilon, p_\epsilon+1} - ab \mathcal{M}_{ab}^{p_\epsilon+1, p_\epsilon+1} \right]. \quad (58)$$

The longitudinal coefficient can be written using the same transverse combination:

$$\begin{aligned} \mathcal{P}_\eta^{(2),L}(\tilde{\nu}) &= \frac{\mathcal{P}_\eta^{(1)}(\tilde{\nu}) + (1/2 + i\eta\tilde{\nu})^2 \mathcal{P}_\eta^{(2),+}(\tilde{\nu})}{\tilde{\nu}^2 + 1/4} \\ &\quad + \frac{2^{2i\eta\tilde{\nu}}}{\tilde{\nu}^2 + 1/4} \text{FP}_{\epsilon=0} \sum_{a,b=\pm 1} \left\{ (1/2 + i\eta\tilde{\nu}) \left[\mathcal{M}_{ab}^{p_\epsilon+2, p_\epsilon} + ib \mathcal{M}_{ab}^{p_\epsilon+2, p_\epsilon+1} \right. \right. \\ &\quad \left. \left. + \mathcal{M}_{ab}^{p_\epsilon, p_\epsilon+2} + ia \mathcal{M}_{ab}^{p_\epsilon+1, p_\epsilon+2} \right] + \mathcal{M}_{ab}^{p_\epsilon+2, p_\epsilon+2} \right\}. \end{aligned} \quad (59)$$

The operator $\text{FP}_{\epsilon=0}$ selects the constant coefficient of the Laurent expansion after assembling the indicated seed sum. In particular, the seed expressions are combined before setting ϵ to zero. The first term proportional to $\mathcal{P}_\eta^{(1)}$ in (59) is the contact contribution in (34).

D Box Polarization Coefficients

D.1 Polarization sums

Let $\lambda_1, \lambda_3 = L, \pm$ denote the polarizations of the hard lines with momenta $\mathbf{k}_1$ and $\mathbf{k}_3$. With $\mathbf{K} = \mathbf{q} + \mathbf{k}_s$, define the soft-polarization sum by

$$\begin{aligned} \sum_{\sigma_1, \sigma_2=L, \pm} [\beta(x)]^{\delta_{\sigma_1 L} + \delta_{\sigma_2 L}} \int \frac{d^3 \mathbf{q}}{(2\pi)^3} \left(\frac{q}{2}\right)^{-2x} \left(\frac{K}{2}\right)^{-2x} (e_{\mathbf{k}_1}^{(\lambda_1)*} \cdot e_{\mathbf{K}}^{(\sigma_1)}) (e_{\mathbf{K}}^{(\sigma_1)*} \cdot e_{\mathbf{k}_3}^{(\lambda_3)}) \\ \times (e_{\mathbf{k}_3}^{(\lambda_3)*} \cdot e_{\mathbf{q}}^{(\sigma_2)}) (e_{\mathbf{q}}^{(\sigma_2)*} \cdot e_{\mathbf{k}_1}^{(\lambda_1)}) = \frac{4^{2x} k_s^{3-4x}}{(4\pi)^{3/2}} \mathcal{I}_{\text{Box}}^{\lambda_1 \lambda_3}(x, \theta_1, \theta_3, \phi_3). \end{aligned} \quad (60)$$

The angles follow (1) and the convention stated below it. The two transverse hard-line coefficients $\mathcal{P}_\eta^{(2),+} = \mathcal{P}_\eta^{(2),-}$ are equal, so it suffices to retain three groups:

$$\mathcal{I}_{\text{Box}}^{LH} \equiv \sum_{\lambda=\pm} \left(\mathcal{I}_{\text{Box}}^{L\lambda} + \mathcal{I}_{\text{Box}}^{\lambda L} \right), \quad \mathcal{I}_{\text{Box}}^{HH} \equiv \sum_{\lambda_1, \lambda_3=\pm} \mathcal{I}_{\text{Box}}^{\lambda_1 \lambda_3}, \quad (61)$$

together with $\mathcal{I}_{\text{Box}}^{LL}$. Here H in a polarization superscript denotes the sum over transverse helicities.

For $x = -i\tilde{\nu}$, the longitudinal result is

$$\begin{aligned} & \mathcal{I}_{\text{Box}}^{LL}(-i\tilde{\nu}, \theta_1, \theta_3, \phi_3) \\ &= -\frac{4^{-3-2i\tilde{\nu}}(2\tilde{\nu} - 3i)\Gamma(-3/2 - 2i\tilde{\nu})\Gamma(1 + 2i\tilde{\nu})\sinh^2(\pi\tilde{\nu})}{\pi(\tilde{\nu} - i)(\tilde{\nu} - 2i)(2\tilde{\nu} - i)} \\ & \times \left\{ -16 - 19i\tilde{\nu} + 6\tilde{\nu}^2 + \tilde{\nu}(2\tilde{\nu} - i) \left[\cos(2\theta_1) + \cos(2\theta_3) + 3\cos(2\theta_1)\cos(2\theta_3) \right] \right. \\ & \quad + 4(2\tilde{\nu}^2 - 5i\tilde{\nu} - 3)\cos(2\phi_3)\sin^2\theta_1\sin^2\theta_3 \\ & \quad \left. - 8\tilde{\nu}(\tilde{\nu} - i)\cos\phi_3\sin(2\theta_1)\sin(2\theta_3) \right\}. \end{aligned} \quad (62)$$

Polarization completeness and (53) reduce the other two groups to

$$\mathcal{I}_{\text{Box}}^{LH} = \frac{2}{3}\mathcal{I}_{\text{Bub}} - 2\mathcal{I}_{\text{Box}}^{LL}, \quad (63)$$

$$\mathcal{I}_{\text{Box}}^{HH} = \frac{1}{3}\mathcal{I}_{\text{Bub}} + \mathcal{I}_{\text{Box}}^{LL}. \quad (64)$$

The arguments are the same as in (62), with $\mathcal{I}_{\text{Bub}}$ evaluated at $-i\tilde{\nu}$. Together with (50), these are explicit Gamma-function expressions for all three groups. They satisfy $\mathcal{I}_{\text{Box}}^{LL} + \mathcal{I}_{\text{Box}}^{LH} + \mathcal{I}_{\text{Box}}^{HH} = \mathcal{I}_{\text{Bub}}$.

The dimensionless box coefficient is consequently

$$\begin{aligned} C^{\text{Box}}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) &= \frac{\Gamma^4(-i\tilde{\nu})}{(4\pi)^{7/2}} \left\{ [\mathcal{P}_+^{(2),L} - \mathcal{P}_+^{(2),+}]^2 \mathcal{I}_{\text{Box}}^{LL} \right. \\ & \quad \left. + \frac{1}{3}\mathcal{P}_+^{(2),+} [2\mathcal{P}_+^{(2),L} + \mathcal{P}_+^{(2),+}] \mathcal{I}_{\text{Bub}} \right\}, \end{aligned} \quad (65)$$

where all time-integral coefficients have argument $\tilde{\nu}$ and are given in App. C. The opposite nonlocal branch follows by complex conjugation.

D.2 Angular projections

The five functions in (40) can be read off from (65), or equivalently obtained from the following projections:

$$A_1(\tilde{\nu}) = \frac{1}{2} \left[C^{\text{Box}}(\tilde{\nu}, \frac{\pi}{2}, \frac{\pi}{2}, 0) + C^{\text{Box}}(\tilde{\nu}, \frac{\pi}{2}, \frac{\pi}{2}, \frac{\pi}{2}) \right], \quad (66)$$

$$A_2(\tilde{\nu}) = C^{\text{Box}}(\tilde{\nu}, \frac{\pi}{2}, 0, 0) - A_1(\tilde{\nu}), \quad (67)$$

$$A_3(\tilde{\nu}) = C^{\text{Box}}(\tilde{\nu}, 0, 0, 0) - A_1(\tilde{\nu}) - 2A_2(\tilde{\nu}), \quad (68)$$

$$B(\tilde{\nu}) = \frac{1}{2} \left[C^{\text{Box}}(\tilde{\nu}, \frac{\pi}{4}, \frac{\pi}{4}, 0) - C^{\text{Box}}(\tilde{\nu}, \frac{\pi}{4}, \frac{\pi}{4}, \pi) \right], \quad (69)$$

$$D(\tilde{\nu}) = \frac{1}{2} \left[C^{\text{Box}}(\tilde{\nu}, \frac{\pi}{2}, \frac{\pi}{2}, 0) - C^{\text{Box}}(\tilde{\nu}, \frac{\pi}{2}, \frac{\pi}{2}, \frac{\pi}{2}) \right]. \quad (70)$$

These projections retain the normalization of the single ordering in (38).

References

- [1] S. Lu, Y. Wang, and Z.-Z. Xianyu, “A Cosmological Higgs Collider,” *Journal of High Energy Physics* **2020** no. 2, (2020) 11, [arXiv:1907.07390 \[hep-th\]](#).
<https://arxiv.org/abs/1907.07390>.
- [2] Z. Qin and Z.-Z. Xianyu, “Helical Inflation Correlators: Partial Mellin-Barnes and Bootstrap Equations,” *Journal of High Energy Physics* **2023** no. 4, (Apr., 2023) 59.
<http://arxiv.org/abs/2208.13790>. arXiv:2208.13790 [astro-ph, physics:hep-ph, physics:hep-th].
- [3] Z. Qin and Z.-Z. Xianyu, “Closed-Form Formulae for Inflation Correlators,” *Journal of High Energy Physics* **2023** no. 7, (July, 2023) 1. <http://arxiv.org/abs/2301.07047>.
arXiv:2301.07047 [hep-th].
- [4] Z. Qin and Z.-Z. Xianyu, “Inflation correlators at the one-loop order: nonanalyticity, factorization, cutting rule, and OPE,” *Journal of High Energy Physics* **2023** no. 9, (Sept., 2023) 116. [https://link.springer.com/10.1007/JHEP09\(2023\)116](https://link.springer.com/10.1007/JHEP09(2023)116).